

# Lecture 2: Univariate Population Inference

August 4, 2023

# Review

- ▶ Last lecture, we discussed description/summarization with means, conditional means, and differences in conditional means.
- ▶ We considered the properties of these summaries (e.g., balance point, least squares, etc.) and how much was left out by summarizing (e.g., squared residuals)
- ▶ This was done in the context where we only cared about the sample, either provisionally or because we had sampled everyone (i.e., a census).

# Today

- ▶ Today we consider the context where we have a random sample from a large population.
- ▶ Our goal is to summarize the population with population means, conditional population means, and differences in conditional population means.
- ▶ Summarizing data we don't have (everything in the population that isn't in the sample) is more difficult than summarizing data we do have (sample).

# Outline



# Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of  $n = 625$  potential Georgia voters, randomly selected (with replacement) from the population of  $N = 7,233,584$  on the public voters register, to ask whether they planned to vote.

# Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of  $n = 625$  potential Georgia voters, randomly selected (with replacement) from the population of  $N = 7,233,584$  on the public voters register, to ask whether they planned to vote.

Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

# Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of  $n = 625$  potential Georgia voters, randomly selected (with replacement) from the population of  $N = 7,233,584$  on the public voters register, to ask whether they planned to vote.

Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

Table: Sample

|       |   |   |   |   |     |     |                 |
|-------|---|---|---|---|-----|-----|-----------------|
| $i$   | 1 | 2 | 3 | 4 | ... | 625 | $\bar{y}_{625}$ |
| $y_i$ | 1 | 1 | 0 | 1 | ... | 0   | .66             |

# Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of  $n = 625$  potential Georgia voters, randomly selected (with replacement) from the population of  $N = 7,233,584$  on the public voters register, to ask whether they planned to vote.

Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

Table: Sample

|       |   |   |   |   |     |     |                 |
|-------|---|---|---|---|-----|-----|-----------------|
| $i$   | 1 | 2 | 3 | 4 | ... | 625 | $\bar{y}_{625}$ |
| $y_i$ | 1 | 1 | 0 | 1 | ... | 0   | .66             |

Remember that the mean of a binary variable is a proportion.



# Sample versus Population

Table: Sample

|       |   |   |   |   |     |     |                 |
|-------|---|---|---|---|-----|-----|-----------------|
| $i$   | 1 | 2 | 3 | 4 | ... | 625 | $\bar{y}_{625}$ |
| $y_i$ | 1 | 1 | 0 | 1 | ... | 0   | .68             |

# Sample versus Population

Table: Sample

|       |   |   |   |   |     |     |                 |
|-------|---|---|---|---|-----|-----|-----------------|
| $i$   | 1 | 2 | 3 | 4 | ... | 625 | $\bar{y}_{625}$ |
| $y_i$ | 1 | 1 | 0 | 1 | ... | 0   | .68             |

After election day, we found that in fact 4, 999, 960 people, or roughly 70% of registered voters, turned out to vote.

Table: Population

|       |   |   |   |   |     |     |     |     |           |                       |
|-------|---|---|---|---|-----|-----|-----|-----|-----------|-----------------------|
| $j$   | 1 | 2 | 3 | 4 | ... | ... | ... | ... | 7,233,584 | $\bar{y}_{7,233,584}$ |
| $y_j$ | 0 | 1 | 0 | 1 | ... | ... | ... | ... | 1         | .70                   |

# Questions

1. .68 is close to .70, did we get lucky?
- 2.
- 3.

# Sampling Distribution of $\bar{Y}_{625}$

| $i$ | $J_1$   | $Y_1$ | $J_2$   | $Y_2$ | $\dots$ | $J_{625}$ | $Y_{625}$ | $\bar{Y}_{625}$ |
|-----|---------|-------|---------|-------|---------|-----------|-----------|-----------------|
|     | 9562350 | 1     | 8763351 | 1     | $\dots$ | 1294801   | 0         | .68             |
|     |         |       |         |       |         |           |           |                 |

## Sampling Distribution of $\bar{Y}_{625}$

| $i$ | 1       | 2       | ...       | 625       |                 |
|-----|---------|---------|-----------|-----------|-----------------|
| $s$ | $J_1$   | $J_2$   | $J_{625}$ | $Y_{625}$ | $\bar{Y}_{625}$ |
| 1   | 9562350 | 8763351 | 1294801   | 0         | .68             |
| 2   | 5331704 | 4533839 | 3342359   | 1         | .70             |
|     |         |         |           |           |                 |

# Sampling Distribution of $\bar{Y}_{625}$

| $i$   | 1       |       | 2        |       | ... | 625       |           | $\bar{Y}_{625}$ |
|-------|---------|-------|----------|-------|-----|-----------|-----------|-----------------|
| $s$   | $J_1$   | $Y_1$ | $J_2$    | $Y_2$ | ... | $J_{625}$ | $Y_{625}$ |                 |
| 1     | 9562350 | 1     | 8763351  | 1     | ... | 1294801   | 0         | .68             |
| 2     | 5331704 | 0     | 4533839  | 1     | ... | 3342359   | 1         | .70             |
| 3     | 5129936 | 0     | 10981600 | 0     | ... | 4096184   | 1         | .75             |
| 4     | 803605  | 0     | 7036389  | 1     | ... | 803605    | 0         | .73             |
| 5     | 148567  | 0     | 3833847  | 1     | ... | 4769869   | 1         | .69             |
| ⋮     | ⋮       | ⋮     | ⋮        | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |
| 1 mil | 4163458 | 0     | 8384613  | 1     | ... | 377981    | 1         | .74             |
| ⋮     | ⋮       | ⋮     | ⋮        | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |
| ⋮     | ⋮       | ⋮     | ⋮        | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |

# Sampling Distribution of $\bar{Y}_{625}$

| $i$        | 1       |       | 2        |       | ... | 625       |           | $\bar{Y}_{625}$ |
|------------|---------|-------|----------|-------|-----|-----------|-----------|-----------------|
| $s$        | $J_1$   | $Y_1$ | $J_2$    | $Y_2$ | ... | $J_{625}$ | $Y_{625}$ |                 |
| 1          | 9562350 | 1     | 8763351  | 1     | ... | 1294801   | 0         | .68             |
| 2          | 5331704 | 0     | 4533839  | 1     | ... | 3342359   | 1         | .70             |
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| ⋮          | ⋮       | ⋮     | ⋮        | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |
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| ⋮          | ⋮       | ⋮     | ⋮        | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |
| $f(\cdot)$ |         |       |          |       |     |           |           |                 |

- What is the distribution of  $Y_1$ ?

# Sampling Distribution of $\bar{Y}_{625}$

| $i$     | 1         |       | 2         |       | ... | 625       |           | $\bar{Y}_{625}$ |
|---------|-----------|-------|-----------|-------|-----|-----------|-----------|-----------------|
| $s$     | $J_1$     | $Y_1$ | $J_2$     | $Y_2$ | ... | $J_{625}$ | $Y_{625}$ |                 |
| 1       | 9562350   | 1     | 8763351   | 1     | ... | 1294801   | 0         | .68             |
| 2       | 5331704   | 0     | 4533839   | 1     | ... | 3342359   | 1         | .70             |
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| ⋮       | ⋮         | ⋮     | ⋮         | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |
| 1 mil   | 4163458   | 0     | 8384613   | 1     | ... | 377981    | 1         | .74             |
| ⋮       | ⋮         | ⋮     | ⋮         | ⋮     | ⋮   | ⋮         | ⋮         | ⋮               |
| $f.(.)$ | Bern(.70) |       | Bern(.70) |       |     | Bern(.70) |           |                 |

- ▶ What is the distribution of  $Y_1$ ?
- ▶ What is the distribution of  $\sum_i^{625} Y_i$ ?



# Sampling Distribution of $\bar{Y}_{625}$

| $i$        | 1         |       | 2         |       | ... | 625       |           | $\bar{Y}_{625}$             |
|------------|-----------|-------|-----------|-------|-----|-----------|-----------|-----------------------------|
| $s$        | $J_1$     | $Y_1$ | $J_2$     | $Y_2$ | ... | $J_{625}$ | $Y_{625}$ |                             |
| 1          | 9562350   | 1     | 8763351   | 1     | ... | 1294801   | 0         | .68                         |
| 2          | 5331704   | 0     | 4533839   | 1     | ... | 3342359   | 1         | .70                         |
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| ⋮          | ⋮         | ⋮     | ⋮         | ⋮     | ⋮   | ⋮         | ⋮         | ⋮                           |
| 1 mil      | 4163458   | 0     | 8384613   | 1     | ... | 377981    | 1         | .74                         |
| ⋮          | ⋮         | ⋮     | ⋮         | ⋮     | ⋮   | ⋮         | ⋮         | ⋮                           |
| $f(\cdot)$ | Bern(.70) |       | Bern(.70) |       |     | Bern(.70) |           | $\frac{Bin(625, .70)}{625}$ |

- ▶ What is the distribution of  $Y_1$ ?
- ▶ What is the distribution of  $\sum_i^{625} Y_i$ ?
- ▶ What two numbers show that we sampled with replacement?

# Review of Bernoulli and Binomial Distributions

We can write distributions for our random variables (RVs) that describe the probability of getting particular values for these variables.

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We can write distributions for our random variables (RVs) that describe the probability of getting particular values for these variables.

$Y_1, \dots, Y_{625}$  are Bernoulli RVs:

$$f_{Y_1}(y_1) = (.71)^{y_1} (.29)^{1-y_1} \text{ for } y_1 = 0, 1$$

$$\vdots = \vdots$$

$$f_{Y_{625}}(y_{625}) = (.71)^{y_{625}} (.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1$$

# Review of Bernoulli and Binomial Distributions

We can write distributions for our random variables (RVs) that describe the probability of getting particular values for these variables.

$Y_1, \dots, Y_{625}$  are Bernoulli RVs:

$$\begin{aligned} f_{Y_1}(y_1) &= (.71)^{y_1} (.29)^{1-y_1} \text{ for } y_1 = 0, 1 \\ \vdots &= \vdots \\ f_{Y_{625}}(y_{625}) &= (.71)^{y_{625}} (.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1 \end{aligned}$$

$\sum_{i=1}^{625} Y_i$  is a Binomial RV:

$$f_{\sum_{i=1}^{625} Y_i} \left( \sum_{i=1}^{625} y_i \right) = \binom{625}{k} (.70)^{\sum_{i=1}^{625} y_i} (.30)^{625 - \sum_{i=1}^{625} y_i} \text{ for } \sum_{i=1}^{625} y_i = 0, \dots, 625$$

# The Sampling Distribution in $\mathbb{R}$

```
# Resample the number of voters 1,000,000  
# times and store these 1,000,000  
# numbers in a vector.
```

# The Sampling Distribution in R

```
# Resample the number of voters 1,000,000  
# times and store these 1,000,000  
# numbers in a vector.  
  
sumY_vec <- rbinom(1000000, size=625, prob=.70)
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# times and store these 1,000,000  
# numbers in a vector.  
  
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# Divide all of these numbers  
# by the sample size.  
  
Ybar_vec <- sumY_vec/625
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# times and store these 1,000,000
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# Divide all of these numbers
# by the sample size.

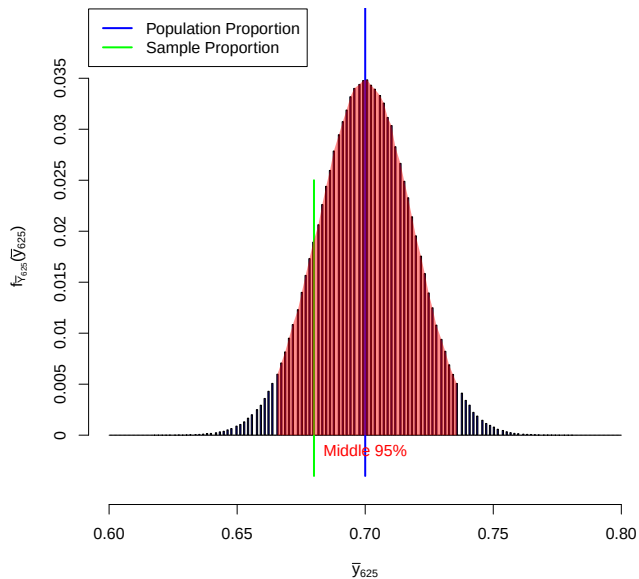
Ybar_vec <- sumY_vec/625

# Plot a histogram

hist(Ybar_vec)
```



# Sampling Distribution of $\bar{Y}_{625}$



# Questions

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1. .68 is close to .70, did we get lucky? No!
2. Could we have predicted how close we would get before seeing the .70?
- 3.

# Bootstrapped Sampling Distributions

At the time of our sample, we don't observe the population or population proportion ( $.70$ ), so we cannot construct the sampling distribution.

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However, we can take repeated random samples *with replacement* of size 625 from the sample of size 625.

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|-------|---|---|---|---|-----|-----|-----------------|
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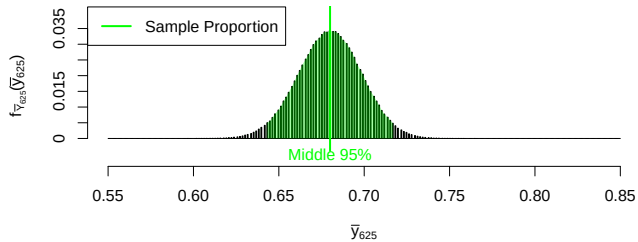
Table: Sample

|       |   |   |   |   |     |     |                 |
|-------|---|---|---|---|-----|-----|-----------------|
| $i$   | 1 | 2 | 3 | 4 | ... | 625 | $\bar{y}_{625}$ |
| $y_i$ | 1 | 1 | 0 | 1 | ... | 0   | .68             |

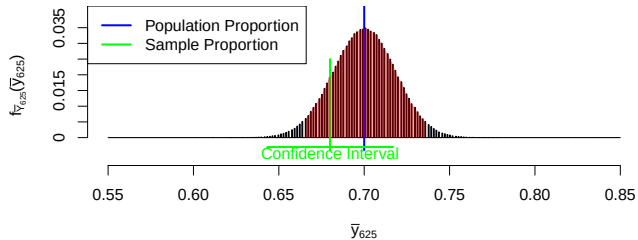
This is equivalent to replacing .70 with .68 in the R code.

```
sumY_vec <- rbinom(1000000, size=625, prob=.68)
```

Estimated Sampling Distribution of  $\bar{Y}_{625}$



Sampling Distribution of  $\bar{Y}_{625}$





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- ▶ the width of the middle 95% of the sampling distribution is equal to the width of the middle 95% of the estimated sampling distribution,

# Informal Conditions for Confidence Intervals

If the following are true,

- ▶ the sampling distribution is symmetric about the population proportion,
- ▶ the estimated sampling distribution is symmetric about the sample proportion,
- ▶ the width of the middle 95% of the sampling distribution is equal to the width of the middle 95% of the estimated sampling distribution,

then the following logic for the 95% confidence interval will hold.

# The Logic of Confidence Intervals

1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.

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2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.

# The Logic of Confidence Intervals

1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.
3. There is a 95% probability of getting a sample proportion in the middle 95% of the sampling distribution, therefore there is 95% probability of getting a 95% confidence interval that covers the population proportion.



What do we mean by a random sample?

| $i$<br>$s$ | 1<br>$Y_1$ | 2<br>$Y_2$ | ...      | 625<br>$Y_{625}$ | $\bar{Y}_{625}$                    |
|------------|------------|------------|----------|------------------|------------------------------------|
| 1          | 1          | 1          | ...      | 0                | .68                                |
| 2          | 0          | 1          | ...      | 1                | .70                                |
| 3          | 0          | 0          | ...      | 1                | .75                                |
| 4          | 0          | 1          | ...      | 1                | .73                                |
| 5          | 0          | 1          | ...      | 1                | .69                                |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$                           |
| 1 mil      | 0          | 1          | ...      | 1                | .74                                |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$                           |
| $f(\cdot)$ | Be(.71)    | Be(.71)    |          | Be(.71)          | $\frac{\text{Bin}(625, .71)}{625}$ |
| E.(.)      | .71        | .71        |          | .71              | .71                                |

## Beyond binary

| $i$      | 1        | 2        | ...      | 625       |                 |
|----------|----------|----------|----------|-----------|-----------------|
| $s$      | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$ |
| 1        | 35       | 50       | ...      | 22        | 45              |
| 2        | 28       | 19       | ...      | 65        | 47              |
| 3        | 33       | 27       | ...      | 45        | 42              |
| 4        | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| 5        | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| 1 mil    | 41       | 53       | ...      | 21        | 46              |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $f.(.)$  | ?        | ?        |          | ?         | ?               |
| $E.(.)$  | 45       | 45       |          | 45        | ?               |

Important properties:

$$E[c] = c$$

$$E[aX] = aE[X]$$

Prove these assuming  $X$  is a discrete RV.

Higher moments about the mean. The  $j$ th *central* moment:

$$\mu_j = \text{E} \left[ (X - \text{E}[X])^j \right] .$$

The 2nd *central* moment, the variance:

$$\begin{aligned} V(X) &= E \left[ (X - E[X])^2 \right] \\ &= \\ &= \\ &= \\ &= \\ &= E \left[ X^2 \right] - E[X]^2 \end{aligned}$$

- ▶ The average squared deviation from the expected value.
- ▶ Expectation of the square minus the square of the expectation.

What does this mean, physically?

The 2nd *central* moment, the variance:

$$\begin{aligned} V(X) &= E \left[ (X - E[X])^2 \right] \\ &= E \left[ X^2 - 2XE[X] + E[X]^2 \right] \\ &= E \left[ X^2 \right] - 2E[XE[X]] + E \left[ E[X]^2 \right] \\ &= E \left[ X^2 \right] - 2E[X]E[X] + E[X]^2 \\ &= E \left[ X^2 \right] - 2E[X]^2 + E[X]^2 \\ &= E \left[ X^2 \right] - E[X]^2 \end{aligned}$$

- ▶ The average squared deviation from the expected value.
- ▶ Expectation of the square minus the square of the expectation.

What does this mean, physically?

What do we mean by random draws?

| $i$        | 1          | 2          | ...      | 625        |                             |
|------------|------------|------------|----------|------------|-----------------------------|
| $s$        | $Y_1$      | $Y_2$      | ...      | $Y_{625}$  | $\bar{Y}_{625}$             |
| 1          | 1          | 1          | ...      | 0          | .68                         |
| 2          | 0          | 1          | ...      | 1          | .70                         |
| 3          | 0          | 0          | ...      | 1          | .75                         |
| 4          | 0          | 1          | ...      | 1          | .73                         |
| 5          | 0          | 1          | ...      | 1          | .69                         |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$   | $\vdots$                    |
| 1 mil      | 0          | 1          | ...      | 1          | .74                         |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$   | $\vdots$                    |
| $f(\cdot)$ | Be(.71)    | Be(.71)    |          | Be(.71)    | $\frac{Bin(625, .71)}{625}$ |
| E.(.)      | .71        | .71        |          | .71        | .71                         |
| V.(.)      | .71(1-.71) | .71(1-.71) |          | .71(1-.71) | ?                           |

## Beyond binary

| $i$        | 1        | 2        | ...      | 625       |                 |
|------------|----------|----------|----------|-----------|-----------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$ |
| 1          | 35       | 50       | ...      | 22        | 45              |
| 2          | 28       | 19       | ...      | 65        | 47              |
| 3          | 33       | 27       | ...      | 45        | 42              |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$        |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| 1 mil      | 41       | 53       | ...      | 21        | 46              |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$        |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?               |
| $V(\cdot)$ | 25       | 25       |          | 25        | ?               |



Important properties:

$$V[c] = 0$$

$$V[aX] = a^2 V[X]$$

$$V[c + aX] = a^2 V[X]$$

Prove these assuming  $X$  is a discrete RV.

What do these properties mean?

| $i$        | 1            | 2            | ...      | 625          |                                 |
|------------|--------------|--------------|----------|--------------|---------------------------------|
| $s$        | $Y_1$        | $Y_2$        | ...      | $Y_{625}$    | $\bar{Y}_{625}$                 |
| 1          | 1            | 1            | ...      | 0            | .68                             |
| 2          | 0            | 1            | ...      | 1            | .70                             |
| 3          | 0            | 0            | ...      | 1            | .75                             |
| 4          | 0            | 1            | ...      | 1            | .73                             |
| 5          | 0            | 1            | ...      | 1            | .69                             |
| $\vdots$   | $\vdots$     | $\vdots$     | $\vdots$ | $\vdots$     | $\vdots$                        |
| 1 mil      | 0            | 1            | ...      | 1            | .74                             |
| $\vdots$   | $\vdots$     | $\vdots$     | $\vdots$ | $\vdots$     | $\vdots$                        |
| $\vdots$   | $\vdots$     | $\vdots$     | $\vdots$ | $\vdots$     | $\vdots$                        |
| $f(\cdot)$ | $Be(.71)$    | $Be(.71)$    |          | $Be(.71)$    | $\frac{Bin(625, .71)}{625}$     |
| $E(\cdot)$ | .71          | .71          |          | .71          | .71                             |
| $V(\cdot)$ | $.71(1-.71)$ | $.71(1-.71)$ |          | $.71(1-.71)$ | $\frac{625(.71)(1-.71)}{625^2}$ |

Definition:

Standard deviation:  $\sigma(X) = \sqrt{V(X)}$ .

Definition: mean squared error (MSE) about a guess  $c$ :

$$\begin{aligned} \mathbb{E} \left[ (X - c)^2 \right] &= \\ &= \\ &= V(X) + (\mathbb{E}[X] - c)^2 \end{aligned}$$

Definition: mean squared error (MSE) about a guess  $c$ :

$$\begin{aligned} E \left[ (X - c)^2 \right] &= E[X^2] - 2cE[X] + c^2 \\ &= E[X^2] - E[X]^2 + E[X]^2 - 2cE[X] + c^2 \\ &= V(X) + (E[X] - c)^2 \end{aligned}$$

What is the minimum MSE choice of  $c$ ?

Defining expectation over a vector of RVs.

$$E[(X, Y)] = (E[X], E[Y])$$

Defining an expectation over functions multiple R.V.s.

Let  $Z = h(X, Y)$  be a scalar function of the random vector  $(X, Y)$  as we proceed.

Discrete:

$$E[Z] = \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f(x, y)$$

Continuous:

$$E[Z] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dy dx$$

So this is just a straightforward generalization (working with joint PMFs/PDFs).

Important property: Prove it for discrete.

$$E[aX + bY + c] =$$

$$=$$
$$=$$
$$+$$

$$= aE[X] + bE[Y] + c$$



Important property: Prove it for discrete.

$$\begin{aligned} E[aX + bY + c] &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f(x, y) \\ &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} (ax + by + c) f(x, y) \\ &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} a \cdot x \cdot f(x, y) \\ &\quad + \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} b \cdot y \cdot f(x, y) + c \\ &= aE[X] + bE[Y] + c \end{aligned}$$

What do these properties mean?

| $i$        | 1        | 2        | ...      | 625       |                                   |
|------------|----------|----------|----------|-----------|-----------------------------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\frac{1}{2}Y_1 + \frac{1}{2}Y_2$ |
| 1          | 35       | 50       | ...      | 22        | $\frac{1}{2}35 + \frac{1}{2}50$   |
| 2          | 28       | 19       | ...      | 65        | $\frac{1}{2}28 + \frac{1}{2}19$   |
| 3          | 33       | 27       | ...      | 45        | $\frac{1}{2}33 + \frac{1}{2}27$   |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$                          |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$                          |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                          |
| 1 mil      | 41       | 53       | ...      | 21        | $\frac{1}{2}41 + \frac{1}{2}53$   |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                          |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$                          |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?                                 |
| $E(\cdot)$ | 45       | 45       |          | 45        | $\frac{1}{2}45 + \frac{1}{2}45$   |

# Understanding i.i.d.

What does i.i.d. stand for:

- ▶ Independent (i.e., observations independent from each other)
- ▶ Identically distributed (i.e., all observations drawn from same distribution)

# Implications of independence

$Y_1$  and  $Y_2$  are *independent* if and only if  
 $f_{Y_1, Y_2}(y_1, y_2) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$ , for all  $(y_1, y_2)$ .

If  $Y_1$  and  $Y_2$  are independent, then:

- Covariance is zero:

$$C(Y_1, Y_2) = 0.$$

- Variances are additive:

$$V(Y_1 + Y_2) = V(Y_1) + V(Y_2).$$

## Implications of identically distributed

$Y_1$  and  $Y_2$  are *independent* if and only if  
 $f_{Y_1, Y_2}(y_1, y_2) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$ , for all  $(y_1, y_2)$ .

$Y_1$  and  $Y_2$  are *identically distributed* if and only if  
 $f_{Y_1}(y_1) = f_{Y_2}(y_2)$  for all  $(y_1, y_2)$ .

So, we have  $n$  random variables that generate  $n$  values.

We can subscript each of these  $n$  random variables, as well as aggregate them.

The random vector  $\mathbf{Y} = (Y_1, \dots, Y_n)'$ .

Under the assumption of *random sampling* from  $F_Y$ , the collection of values that you see are each produced by an identical random process.

The actual values that we observe are denoted with  $\mathbf{y} = (y_1, \dots, y_n)'$ .

For now, we focus on learning about the population mean ( $E[Y] = \mu$ ), although to do so, we will need to also learn about the nuisance parameter ( $V[Y]$ ).

We want an estimator with small mean squared error (recall that this is equivalent to squared bias plus variance) and an estimator that allows the production of tight confidence intervals.

A sample statistic *summarizes* those  $n$  values. When used to estimate a population parameter, also considered an estimator.

Sample mean:

$$\hat{\mu}_{\text{sample mean}} = \bar{Y} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{1}{n} (Y_1 + \dots + Y_n).$$

Class of linear combinations:

$$\hat{\mu}_{\mathbf{c}} = \sum_{i=1}^n c_i Y_i = (c_1 Y_1 + \dots + c_n Y_n).$$



Which column is the sampling distribution of the sample mean

| $i$<br>$s$ | 1<br>$Y_1$ | 2<br>$Y_2$ | ...      | 625<br>$Y_{625}$ | $\bar{Y}_{625}$ |
|------------|------------|------------|----------|------------------|-----------------|
| 1          | 35         | 50         | ...      | 22               | 45              |
| 2          | 28         | 19         | ...      | 65               | 47              |
| 3          | 33         | 27         | ...      | 45               | 42              |
| 4          | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| 5          | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| 1 mil      | 41         | 53         | ...      | 21               | 46              |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| $f.(.)$    | ?          | ?          |          | ?                | ?               |
| $E.(.)$    | 45         | 45         |          | 45               | ?               |
| $V.(.)$    | 25         | 25         |          | 25               | ?               |

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean,  $E[\bar{Y}] = E[Y]$ .

Proof:

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{1}{n}(Y_1 + \dots + Y_n)\right] \\ &= \\ &= \\ &= \\ &= \\ &= E[Y_1]. \end{aligned}$$

The sample mean is unbiased

| $i$<br>$s$ | 1<br>$Y_1$ | 2<br>$Y_2$ | ...      | 625<br>$Y_{625}$ | $\bar{Y}_{625}$ |
|------------|------------|------------|----------|------------------|-----------------|
| 1          | 35         | 50         | ...      | 22               | 45              |
| 2          | 28         | 19         | ...      | 65               | 47              |
| 3          | 33         | 27         | ...      | 45               | 42              |
| 4          | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| 5          | $\vdots$   | $\vdots$   | ...      | $\vdots$         | $\vdots$        |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| 1 mil      | 41         | 53         | ...      | 21               | 46              |
| $\vdots$   | $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$         | $\vdots$        |
| $f.(.)$    | ?          | ?          |          | ?                | ?               |
| E.(.)      | 45         | 45         |          | 45               | 45              |
| V.(.)      | 25         | 25         |          | 25               | ?               |

Within the class of linear combinations, what must be true for unbiasedness?

$$\hat{\mu}_{\mathbf{c}} = \sum_{i=1}^n c_i Y_i = (c_1 Y_1 + \dots + c_n Y_n) .$$

Property #2: The *sampling variance* of the sample mean,

$$V[\bar{Y}] = \frac{V[Y]}{n}.$$

Proof:

$$\begin{aligned} V[\bar{Y}] &= V \left[ \left( \frac{1}{n} Y_1 + \dots + \frac{1}{n} Y_n \right) \right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{1}{n} V[Y]. \end{aligned}$$

## Sampling distribution of the sample mean

| $i$        | 1        | 2        | ...      | 625       |                  |
|------------|----------|----------|----------|-----------|------------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$  |
| 1          | 35       | 50       | ...      | 22        | 45               |
| 2          | 28       | 19       | ...      | 65        | 47               |
| 3          | 33       | 27       | ...      | 45        | 42               |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| 1 mil      | 41       | 53       | ...      | 21        | 46               |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?                |
| $E(\cdot)$ | 45       | 45       |          | 45        | 45               |
| $V(\cdot)$ | 25       | 25       |          | 25        | $\frac{25}{625}$ |

Within the class of linear combinations, what is the variance in terms of  $\mathbf{c}$ ?

$$\hat{\mu}_{\mathbf{c}} = \sum_{i=1}^n c_i Y_i = (c_1 Y_1 + \dots + c_n Y_n).$$

What does this expression tell us about minimizing variance?

# Learning about population variance

- ▶ We've shown that the sample mean is unbiased
- ▶ We've shown that the variance of the sample mean is  $V[Y]/n$
- ▶ We still need to learn something about  $V[Y]$



Recall that the “population” variance of a random variable  $Y$ :

$$V(Y) = E[Y^2] - E[Y]^2$$

Can we propose an estimator of  $V(Y)$ ?

Put another way:

- ▶ Can we propose an estimator of  $E[Y^2]$ ?
- ▶ Can we propose an estimator of  $E[Y]^2$ ?

- ▶ Can we propose an estimator of  $E[Y^2]$ ?

- ▶ Plug-in estimator:

$$\overline{Y^2} = \frac{\sum_{i=1}^n Y_i^2}{n}.$$

- ▶  $E[\overline{Y^2}] = E[Y^2]$  (just let  $Y^2 = Z$  and it is just  $E[\overline{Z}] = E[Z]$ ).

- ▶ Can we propose an estimator of  $E[Y]^2$ ?
  - ▶ Plug-in estimator:

$$\bar{Y}^2 = \left[ \frac{\sum_{i=1}^n Y_i}{n} \right]^2.$$

- ▶ What is its expectation ( $E[\bar{Y}^2]$ )?
- ▶ Recall that  $V[\bar{Y}] = E[\bar{Y}^2] - E[\bar{Y}]^2$
- ▶ Simplifies to  $\frac{V[Y]}{n} = E[\bar{Y}^2] - E[\bar{Y}]^2$
- ▶ Rearranging shows that  $E[\bar{Y}^2] = E[\bar{Y}]^2 + \frac{V[Y]}{n}$

Therefore, the plug-in estimator of variance,

$$\overline{Y^2} - \overline{Y}^2.$$

is biased:

$$\begin{aligned} E[\overline{Y^2} - \overline{Y}^2] &= E[Y^2] - (E[Y]^2 + \frac{V[Y]}{n}) \\ &= (E[Y^2] - E[Y]^2) - \frac{V[Y]}{n} \\ &= V(Y) - \frac{V[Y]}{n} \\ &= (1 - \frac{1}{n})V(Y) \\ &= \frac{n-1}{n}V(Y) \end{aligned}$$

However, the bias can be corrected by multiplying the estimator by  $\frac{n}{n-1}$ :

$$\hat{V}(Y) = \frac{n}{n-1} \left[ \overline{Y^2} - \overline{Y}^2 \right].$$

with

$$\blacktriangleright E[\hat{V}(Y)] = \frac{n}{n-1} E \left[ \overline{Y^2} - \overline{Y}^2 \right] = \frac{n}{n-1} \frac{n-1}{n} V(Y)$$

This is what R computes when you type `var ( . )`.

Sampling distribution of the sample variance (fill in for biased sample variance)

| $i$        | 1     | 2     | ... | 625       |                  |              |                              |
|------------|-------|-------|-----|-----------|------------------|--------------|------------------------------|
| $s$        | $Y_1$ | $Y_2$ | ... | $Y_{625}$ | $\bar{Y}_{625}$  | $\hat{V}(Y)$ | $\overline{Y^2} - \bar{Y}^2$ |
| 1          | 35    | 50    | ... | 22        | 45               | 20           |                              |
| 2          | 28    | 19    | ... | 65        | 47               | 17           |                              |
| 3          | 33    | 27    | ... | 45        | 42               | 38           |                              |
| 4          | :     | :     | ... | :         | :                |              |                              |
| 5          | :     | :     | ... | :         | :                |              |                              |
| :          | :     | :     | :   | :         | :                |              |                              |
| :          | :     | :     | :   | :         | :                |              |                              |
| 1 mil      | 41    | 53    | ... | 21        | 46               | 32           |                              |
| :          | :     | :     | :   | :         | :                |              |                              |
| :          | :     | :     | :   | :         | :                |              |                              |
| $f(\cdot)$ | ?     | ?     |     | ?         | ?                |              |                              |
| $E(\cdot)$ | 45    | 45    |     | 45        | 45               | 25           |                              |
| $V(\cdot)$ | 25    | 25    |     | 25        | $\frac{25}{625}$ |              |                              |



# Summary of “Small” Sample Properties

- ▶  $\text{MSE} = E[(\hat{\mu} - \mu)^2]$
- ▶  $\text{Bias} = E[(\hat{\mu} - \mu)]$
- ▶  $\text{Variance} = E[(\hat{\mu} - E[\hat{\mu}])^2]$

$$\underbrace{E[(\hat{\mu} - \mu)^2]}_{\text{MSE}} = \underbrace{E[(\hat{\mu} - \mu)]^2}_{\text{Bias squared}} + \underbrace{E[(\hat{\mu} - E[\hat{\mu}])^2]}_{\text{Variance}}$$

Oct. 4: Asymptotic (i.e., large sample) results and confidence intervals

# Asymptotic properties of estimators

- ▶ Consistency (estimator converges in probability to the population parameter).
- ▶  $MSE \rightarrow 0$  (stronger than consistency, but more intuitive)
- ▶ Asymptotic normality (sampling distribution converges in distribution to normal distribution)

Property # 3: Given finite  $E[Y]$  and finite  $V(Y) > 0$ , then for any  $k > 0$ ,

$$\Pr(|\bar{Y} - E[Y]| \geq k) \leq \frac{V[Y]}{k^2 n}.$$

Proof follows directly from Chebyshev's inequality.

Property # 4. [Stated loosely.] *The sample mean is consistent for the population mean.* Given finite  $E[Y]$  and finite  $V(Y) > 0$ , then as  $n \rightarrow \infty$ ,

$$\Pr(|\bar{Y} - E[Y]| \geq k) \rightarrow 0.$$

Or, equivalently,

$$\bar{Y} \xrightarrow{p} E[Y],$$

where  $\xrightarrow{p}$  is read as “convergence in probability.” This is a form of the *weak law of large numbers* (WLLN).

# Consistency of $\overline{Y}$ for $E[Y]$

The WLLN states that

$$\overline{Y} \xrightarrow{P} E[Y],$$

If we think of  $E[Y]$  as  $\mu$  and  $\overline{Y}$  as  $\hat{\mu}$ , then this is equivalent to saying that  $\hat{\mu}$  is consistent for  $\mu$

## Easier version based on MSE

$$MSE \rightarrow 0 \implies \hat{\theta} \rightarrow_p \theta$$

Implies that it is sufficient to show that Bias and Variance go to zero.

# Consistency

With the person next to you, explain how you know that

$$\overline{Y} \rightarrow_p \mu$$



## Another useful property

(A consequence of the) Continuous Mapping Theorem, stated loosely. Continuous functions are limit-preserving.

For continuous  $f(\cdot)$ , if  $Y \xrightarrow{P} c$ , then  $f(Y) \xrightarrow{P} f(c)$ .

Additionally, for continuous  $f(b, c)$  if  $X \xrightarrow{P} b$  and  $Y \xrightarrow{P} c$ , then  $f(X, Y) \xrightarrow{P} f(b, c)$

# Consistency

With the person next to you, are the following consistent?

- ▶  $\overline{Y^2} - \overline{Y}^2$  for  $\sigma^2$
- ▶  $\hat{V}[Y] = \frac{n}{n-1}(\overline{Y^2} - \overline{Y}^2)$  for  $\sigma^2$

Explain why?

# Inconsistent Examples

With the person next to you, come up with at least one example of an estimator for  $\mu$  that has the following properties:

- ▶ unbiased for  $\mu$  but inconsistent
- ▶ variance goes to zero, but inconsistent for  $\mu$

## Confidence Intervals

Important definition. The *standard error* of the sample mean:

$$\sqrt{V(\bar{Y})}.$$

(This is literally the standard deviation of the sampling distribution of the sample mean.)

An *estimate* of the standard error of the sample mean:

$$\sqrt{\hat{V}(\bar{Y})}.$$

Not unbiased (due to Jensen's inequality), though it is "consistent."

When you see "standard error" in a paper, this is what they mean (an estimate thereof). In theory, it is possible to estimate the sampling variance of the standard error of the sample mean, though.

What is the standard error of the sample mean and what is the estimated standard error of the sample mean?

| $i$        | 1        | 2        | ...      | 625       |                  |              |
|------------|----------|----------|----------|-----------|------------------|--------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$  | $\hat{V}(Y)$ |
| 1          | 35       | 50       | ...      | 22        | 45               | 20           |
| 2          | 28       | 19       | ...      | 65        | 47               | 17           |
| 3          | 33       | 27       | ...      | 45        | 42               | 38           |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         |              |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         |              |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |              |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |              |
| 1 mil      | 41       | 53       | ...      | 21        | 46               | 32           |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |              |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         |              |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?                |              |
| $E(\cdot)$ | 45       | 45       |          | 45        | 45               | 25           |
| $V(\cdot)$ | 25       | 25       |          | 25        | $\frac{25}{625}$ |              |

It is therefore possible to estimate the average of a random variable (the population mean) and estimate our uncertainty (using an estimated SE).

However, we may want to know how likely it would be (across different random samples) for an interval of  $\bar{Y} \pm 2$  SEs to contain  $E[Y]$ .

The central limit theorem provides an approach.

The central limit theorem (CLT) for the sample mean.

In words: if  $n$  is large, the sampling distribution of the sample mean ( $\bar{Y}$ ) is approximately normal, regardless of the distribution of  $Y$ .

In sampling  $n$  times from any R.V.  $Y$  with finite  $E[Y] = \mu$  and finite  $V(Y) = \sigma^2$ , then the standardized sample mean

$$Z = \sqrt{n}(\bar{Y} - \mu)/\sigma$$

converges in distribution as  $n \rightarrow \infty$  to

$$\mathcal{N}(0, 1),$$

where  $\mathcal{N}$  denotes the normal distribution.



I am not going to offer a proof here. The simplest “almost-proof” is given by the de Moivre–Laplace theorem (see the Wikipedia page).

A confidence interval for  $E[Y]$ , loosely speaking, is an *interval estimate* that covers the true  $E[Y]$  with a given probability.

I.e., Suppose we were to repeatedly draw from  $F_{\overline{Y}}$  and compute a (valid) 95% confidence interval for  $E[Y]$ . 95% percent of the time, this interval will encompass  $E[Y]$ .

Suppose we have:

- ▶ An asymptotically normal estimator of  $E[Y]$ :  $\bar{Y}$
- ▶ A consistent estimator of  $nV(\bar{Y})$ :  $n\hat{V}(\bar{Y})$

An approximate 95% confidence interval for  $E[Y]$  would be:

$$\left( \bar{Y} - 1.96\sqrt{\hat{V}(\bar{Y})}, \bar{Y} + 1.96\sqrt{\hat{V}(\bar{Y})} \right).$$

An approximate 95% confidence interval for  $E[Y]$  would be:

$$\left( \bar{Y} - 1.96\sqrt{\frac{\hat{V}(Y)}{n}}, \bar{Y} + 1.96\sqrt{\frac{\hat{V}(Y)}{n}} \right).$$

| $i$     | 1     | 2     | ... | 625       |                  | $MoE$                                      | 95% CI                  |
|---------|-------|-------|-----|-----------|------------------|--------------------------------------------|-------------------------|
| $s$     | $Y_1$ | $Y_2$ | ... | $Y_{625}$ | $\bar{Y}_{625}$  | $1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$ | $\bar{Y}_{625} \pm MoE$ |
| 1       | 35    | 50    | ... | 22        | 45               |                                            |                         |
| 2       | 28    | 19    | ... | 65        | 47               |                                            |                         |
| 3       | 33    | 27    | ... | 45        | 42               |                                            |                         |
| 4       | :     | :     | ... | :         | :                |                                            |                         |
| 5       | :     | :     | ... | :         | :                |                                            |                         |
| :       | :     | :     | :   | :         | :                |                                            |                         |
| 1 mil   | 41    | 53    | ... | 21        | 46               |                                            |                         |
| :       | :     | :     | :   | :         | :                |                                            |                         |
| :       | :     | :     | :   | :         | :                |                                            |                         |
| $f.(.)$ | ?     | ?     |     | ?         | ?                |                                            |                         |
| $E.(.)$ | 45    | 45    |     | 45        | 45               |                                            |                         |
| $V.(.)$ | 25    | 25    |     | 25        | $\frac{25}{625}$ |                                            |                         |

**Table:** Sampling Distributions for  $\bar{Y}_{625}$ , the Margin-of-Error, and the 95% Confidence Interval

| $i$     | 1     | 2     | ... | 625       |                  | $MoE$                                      | 95% CI                  |
|---------|-------|-------|-----|-----------|------------------|--------------------------------------------|-------------------------|
| $s$     | $Y_1$ | $Y_2$ | ... | $Y_{625}$ | $\bar{Y}_{625}$  | $1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$ | $\bar{Y}_{625} \pm MoE$ |
| 1       | 35    | 50    | ... | 22        | 45               | .198                                       |                         |
| 2       | 28    | 19    | ... | 65        | 47               | .195                                       |                         |
| 3       | 33    | 27    | ... | 45        | 42               | .204                                       |                         |
| 4       | :     | :     | ... | :         | :                | .201                                       |                         |
| 5       | :     | :     | ... | :         | :                | .205                                       |                         |
| :       | :     | :     | :   | :         | :                | :                                          |                         |
| 1 mil   | 41    | 53    | ... | 21        | 46               | .197                                       |                         |
| :       | :     | :     | :   | :         | :                | :                                          |                         |
| :       | :     | :     | :   | :         | :                | :                                          |                         |
| $f.(.)$ | ?     | ?     |     | ?         | ?                |                                            |                         |
| $E.(.)$ | 45    | 45    |     | 45        | 45               |                                            |                         |
| $V.(.)$ | 25    | 25    |     | 25        | $\frac{25}{625}$ |                                            |                         |

**Table:** Sampling Distributions for  $\bar{Y}_{625}$ , the Margin-of-Error, and the 95% Confidence Interval

| $i$        | 1        | 2        | ...      | 625       |                  | $MoE$                                      | 95% CI                  |
|------------|----------|----------|----------|-----------|------------------|--------------------------------------------|-------------------------|
| $s$        | $Y_1$    | $Y_2$    | ...      | $Y_{625}$ | $\bar{Y}_{625}$  | $1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$ | $\bar{Y}_{625} \pm MoE$ |
| 1          | 35       | 50       | ...      | 22        | 45               | .198                                       | [44.83, 45.60]          |
| 2          | 28       | 19       | ...      | 65        | 47               | .195                                       | [44.46, 45.22]          |
| 3          | 33       | 27       | ...      | 45        | 42               | .204                                       | [44.61, 45.41]          |
| 4          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         | .201                                       | [45.11, 45.88]          |
| 5          | $\vdots$ | $\vdots$ | ...      | $\vdots$  | $\vdots$         | .205                                       | [44.54, 45.31]          |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         | $\vdots$                                   | $\vdots$                |
| 1 mil      | 41       | 53       | ...      | 21        | 46               | .197                                       | [44.64, 45.46]          |
| $\vdots$   | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$         | $\vdots$                                   | $\vdots$                |
| $f(\cdot)$ | ?        | ?        |          | ?         | ?                |                                            |                         |
| E.(.)      | 45       | 45       |          | 45        | 45               |                                            |                         |
| V.(.)      | 25       | 25       |          | 25        | $\frac{25}{625}$ |                                            |                         |

**Table:** Sampling Distributions for  $\bar{Y}_{625}$ , the Margin-of-Error, and the 95% Confidence Interval



Suppose  $n$  is large.

Define the R.V.

$$Z = \frac{\bar{Y} - E[Y]}{\sqrt{V(\bar{Y})}},$$

where  $Z$  is distributed  $\mathcal{N}(0, 1)$  (by asymptotic normality).

Plugging in the *estimated* standard error.

Define the R.V.

$$Z' = \frac{\bar{Y} - E[Y]}{\sqrt{V(\bar{Y})}} \frac{\sqrt{nV(\bar{Y})}}{\sqrt{n\hat{V}(\bar{Y})}} = \frac{\bar{Y} - E[Y]}{\sqrt{\hat{V}(\bar{Y})}}$$

where  $Z'$  is distributed  $\mathcal{N}(0, 1)$  (by asymptotic normality, the consistency of variance estimator, and Slutsky's Theorem).

## Bootstrapping

1. Take a sample of size  $n$  with replacement from your sample
2. Calculate  $\mu^{(b=1)}$  from that bootstrap sample
3. Repeat for  $b = 1, \dots, B$  where  $B$  is big
4. Do one of the following:
  - ▶ Calculate bootstrapped variance  $\frac{1}{B-1} \sum_{b=1}^B (\mu^{(b)} - \overline{\mu^{(b)}})^2$
  - ▶ Put  $\{\mu^{(b)} : b = 1, \dots, B\}$  in order and report the  $\alpha/2$  and  $1 - \alpha/2$  percentiles
  - ▶ ...

Go to bootstrapping code.

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- ▶ Therefore, the empirical CDF converges to population CDF ( $F(y)$ ) for all  $y$  (AM pp. 116-117)
- ▶ If the empirical CDF is close enough to the population CDF, then a resample with replacement from the sample is like a sample from the population

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- ▶ If the functional is well behaved (e.g., the mean) and not sensitive to small changes in the CDF (e.g., the max is poorly behaved), then the plug-in estimator will converge to the population parameter in many ways (AM pg. 120)
- ▶ And the plug-in estimators from resamples with replacement will converge toward a normal distribution in the same manner (with enough sample size).